

MAT 324 HW02

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Problem 1. Let X, Y be sets, $\mathcal{S} \subseteq 2^X$ and $\mathcal{T} \subseteq 2^Y$ be σ -algebras on X and Y , respectively, and $f: X \rightarrow Y$ be a map. Suppose $\mathcal{T}$ is the σ -algebra generated by some subcollection $\mathcal{T}_0 \subseteq \mathcal{T}$ and $f^{-1}(B) \in \mathcal{S}$ for every $B \in \mathcal{T}_0$. Show that $f^{-1}(B) \in \mathcal{S}$ for every $B \in \mathcal{T}$.

Solution. Let $\mathcal{A} = \{A \subseteq Y : f^{-1}(A) \in \mathcal{S}\}$. Then $\mathcal{A}$ is a σ -algebra on Y because (1) $f^{-1}(\emptyset) = \emptyset \in \mathcal{S}$, so $\emptyset \in \mathcal{A}$; (2) if $A \in \mathcal{A}$ then $f^{-1}(A) \in \mathcal{S}$ hence $X \setminus f^{-1}(A) = f^{-1}(Y \setminus A) \in \mathcal{S}$, so $Y \setminus A \in \mathcal{A}$; and (3) if $A_1, A_2, \dots \in \mathcal{A}$ then $f^{-1}(A_1), f^{-1}(A_2), \dots \in \mathcal{S}$ hence $\bigcup_{n=1}^{\infty} f^{-1}(A_n) = f^{-1}(\bigcup_{n=1}^{\infty} A_n) \in \mathcal{S}$, so $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$. By assumption $\mathcal{T}_0 \subseteq \mathcal{A}$, hence the σ -algebra generated by $\mathcal{T}_0$, namely $\mathcal{T}$, is also contained in $\mathcal{A}$, i.e. $f^{-1}(B) \in \mathcal{S}$ for each $B \in \mathcal{T}$. $\square$

Problem 2. Suppose $A \subseteq B \subseteq C$ are such that A, C are Lebesgue and $|A| = |C| < \infty$. Show that B is also Lebesgue.

Solution. By monotonicity $|A| = |B| = |C|$. Since $|A|$ and $|C|$ are finite, $|C \setminus A| = 0$, so $|B \setminus A| \leq |C \setminus A| = 0$ implies that $B \setminus A$ is Lebesgue because it has outer measure zero. Then B is Lebesgue because it is the union of Lebesgue sets A and $B \setminus A$. $\square$

Problem 3. Let $A \subseteq \mathbb{R}$ with $|A| \neq 0$. Show that there exists $A' \subseteq A$ which is not Lebesgue measurable.

Solution. There exists an integer N such that $|A \cap [N-1, N+1]| > 0$; if such an N didn't exist (so $|A \cap [n-1, n+1]| = 0$ for all $n \in \mathbb{Z}$), then by subadditivity

$$|A| \leq \sum_{n \in \mathbb{Z}} |A \cap [n-1, n+1]| = 0.$$

Let $B = -N + (A \cap [N, N+1]) \subseteq [-1, 1]$, so $|B| > 0$.

The solution now proceeds nearly identically to Axler 2.18. Given numbers x and y in B , say that $x \sim y$ if $x - y \in \mathbb{Q}$. This partitions B into equivalence classes, and we write $[x]$ for the equivalence class of $x \in B$. Define a set S by taking one element from each of these equivalence class, i.e. $S \cap [x]$ has exactly one element for each $x \in B$. If $q_1, q_2, \dots$ is an enumeration of $[-2, 2] \cap \mathbb{Q}$, we have

$$B \subseteq \bigcup_{k=1}^{\infty} (q_k + S), \implies |B| \leq \sum_{k=1}^{\infty} |q_k + S| = \sum_{k=1}^{\infty} |S|,$$

implying $|S| > 0$. On the other hand the translates $q_k + S$ are pairwise disjoint and

$$\bigcup_{k=1}^{\infty} (q_k + S) \subseteq [-3, 3].$$

Since the left-hand side is a countable union of pairwise disjoint sets, if S were Lebesgue then we should be able to write

$$\sum_{k=1}^{\infty} |S| = \sum_{k=1}^{\infty} |q_k + S| = \left| \bigcup_{k=1}^{\infty} (q_k + S) \right| \leq |[-3, 3]| = 6,$$

but this is a contradiction because $|S| > 0$ implies the left-hand side is ∞ , so S is not Lebesgue. $\square$

Problem 4. Let $A \subseteq \mathbb{R}$ and $A^c := \mathbb{R} \setminus A$ be the complement of A . Show that A is Lebesgue if and only if

$$|E| = |E \cap A| + |E \cap A^c| \text{ for all } E \subseteq \mathbb{R}.$$

Solution. Say that $A \subseteq \mathbb{R}$ has the *splitting property* if

$$|E| = |E \cap A| + |E \cap A^c| \text{ for all } E \subseteq \mathbb{R}.$$

We first prove the *only if* direction. Let A be Lebesgue. It suffices to prove

$$|E| \geq |E \cap A| + |E \cap A^c| \text{ for all } E \subseteq \mathbb{R}$$

because the reverse inequality holds by subadditivity. Since $|E|$ is the infimum of $|U|$ over all open sets U (manifested as a union of countably many open intervals) that contain E , for a given $\varepsilon > 0$ we can find an open $U_\varepsilon \subseteq \mathbb{R}$ such that $|U_\varepsilon| \leq |E| + \varepsilon$. This gives

$$|E| \geq |U_\varepsilon| - \varepsilon = |U_\varepsilon \cap A| + |U_\varepsilon \cap A^c| - \varepsilon \geq |E \cap A| + |E \cap A^c| - \varepsilon$$

for all $\varepsilon > 0$, where $|U_\varepsilon| = |U_\varepsilon \cap A| + |U_\varepsilon \cap A^c|$ because the sets $U_\varepsilon \cap A$ and $U_\varepsilon \cap A^c$ are disjoint and Lebesgue. This implies that A has the splitting property if it is Lebesgue.

Suppose conversely that A has the splitting property. If $|A| < \infty$, then by definition of outer measure, given $\varepsilon > 0$, we can find an open set U_ε for which $|U_\varepsilon| < |A| + \varepsilon$. Setting $E = U_\varepsilon$ yields

$$|U_\varepsilon| = |U_\varepsilon \cap A| + |U_\varepsilon \cap A^c| = |A| + |U_\varepsilon \setminus A| \iff |U_\varepsilon \setminus A| = |U_\varepsilon| - |A| < \varepsilon.$$

The existence of such a U_ε is one of the criteria we proved in class for A to be Lebesgue, sets of finite outer measure with the splitting property are Lebesgue.

To finish, suppose that A has the splitting property and $|A| = \infty$. We have proven above that $I_n := [n, n+1]$ has the splitting property because it is Lebesgue, so

$$\begin{aligned} |E \cap A \cap I_n| + |E \cap (A \cap I_n)^c| &= |E \cap A \cap I_n| + |E \cap (A \cap I_n)^c \cap A| + |E \cap (A \cap I_n)^c \cap A^c| \\ &= |E \cap A \cap I_n| + |E \cap A \cap I_n^c| + |E \cap A^c| \\ &= |E \cap A| + |E \cap A^c| = |E|, \end{aligned}$$

which proves that $A \cap I_n$ also has the splitting property. Since $A \cap I_n$ has finite outer measure and has the splitting property, it is Lebesgue, hence so is $A = \bigcup_{n \in \mathbb{Z}} (A \cap I_n)$. $\square$

Problem 5. Let X be a set, $\mathcal{A} \subseteq 2^X$, and $\ell: \mathcal{A} \rightarrow [0, \infty]$ be a function such that $\emptyset \in \mathcal{A}$ and $\ell(\emptyset) = 0$. For each $A \subseteq X$, define

$$\mathcal{L}_A := \left\{ \sum_{n=1}^{\infty} \ell(I_n) : I_1, I_2, \dots \in \mathcal{A}, A \subseteq \bigcup_{n=1}^{\infty} I_n \right\} \subseteq [0, \infty], \quad m(A) = \inf \mathcal{L}_A \in [0, \infty],$$

$$\mathcal{M} = \{A \subseteq X : m(E) = m(E \cap A) + m(E \cap A^c) \text{ for all } E \subseteq X\}. \quad (1)$$

(a) Show that

$$m(\emptyset) = 0; \quad m(A) \leq m(B) \text{ if } A \subseteq B; \quad m\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} m(A_n) \text{ for all } A_1, A_2, \dots \subseteq X. \quad (2)$$

(b) Suppose in addition that $\mathcal{A}$ is an algebra on X and

$$\ell\left(\bigcup_{n=1}^{\infty} I_n\right) = \sum_{n=1}^{\infty} \ell(I_n)$$

for all pairwise disjoint sequences of sets $I_1, I_2, \dots \in \mathcal{A}$ such that $\bigcup_{n=1}^{\infty} I_n \in \mathcal{A}$. Show that $m|_{\mathcal{A}} = \ell$ and $\mathcal{A} \subseteq \mathcal{M}$.

Solution. (a) $\mathcal{L}_{\emptyset}$ contains 0, as we can cover $\emptyset$ with the sequence of sets $I_1 = I_2 = \dots = \emptyset$, so $\inf \mathcal{L}_{\emptyset} = m(\emptyset) \leq 0$, but of course 0 is a lower bound for $\mathcal{L}_{\emptyset}$, so in fact $m(\emptyset) = 0$.

Then, if $A \subseteq B$, a sequence of sets $I_1, I_2, \dots \in \mathcal{A}$ whose union contains B must also contain A , meaning $\mathcal{L}_B \subseteq \mathcal{L}_A$; taking infima translates this to

$$m(A) = \inf \mathcal{L}_A \leq \inf \mathcal{L}_B = m(B).$$

Finally, let $A_1, A_2, \dots \subseteq X$ be a sequence of subsets of X . For each $\varepsilon > 0$ and $k \in \mathbb{N}$, since $m(A_k) = \inf \mathcal{L}_{A_k}$, there exist sets $I_1^{(k)}, I_2^{(k)}, \dots \in \mathcal{A}$ such that $A_k \subseteq \bigcup_{n=1}^{\infty} I_n^{(k)}$ and

$$\sum_{n=1}^{\infty} \ell(I_n^{(k)}) \leq m(A_k) + \frac{\varepsilon}{2^k}.$$

Summing over all k , we get

$$\varepsilon + \sum_{k=1}^{\infty} m(A_k) = \sum_{k=1}^{\infty} \left(m(A_k) + \frac{\varepsilon}{2^k} \right) \geq \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} \ell(I_n^{(k)}) \geq m\left(\bigcup_{k=1}^{\infty} A_k\right)$$

where the last inequality is a result of the fact that the intervals $\{I_n^{(k)}\}_{k,n \in \mathbb{N}}$ cover $\bigcup_{k=1}^{\infty} A_k$.

(b) Suppose that A is in $\mathcal{A}$ and let $\varepsilon > 0$. Then, there exists a sequence $I_1, I_2, \dots \in \mathcal{A}$ such that $A \subseteq \bigcup_{n=1}^{\infty} I_n$ and

$$\sum_{n=1}^{\infty} \ell(I_n) \leq m(A) + \varepsilon.$$

Let $I'_n = I_n \cap A \in \mathcal{A}$ and define $J_1 = I'_1$ and $J_n = I'_n \setminus \bigcup_{k=1}^{n-1} I'_k \in \mathcal{A}$ for $n \geq 2$. The J_n are pairwise disjoint and $\bigcup_{n=1}^{\infty} J_n = A \in \mathcal{A}$, so

$$\ell(A) = \sum_{n=1}^{\infty} \ell(J_n) \leq \sum_{n=1}^{\infty} \ell(I'_n) \leq \sum_{n=1}^{\infty} \ell(I_n) \leq m(A) + \varepsilon.$$

This holds for all $\varepsilon > 0$, implying $\ell(A) \leq m(A)$. On the other hand, A can be covered by A and countably many copies of $\emptyset$, implying that

$$m(A) \leq \ell(A) + \ell(\emptyset) + \ell(\emptyset) + \cdots = \ell(A).$$

Therefore, $m(A) = \ell(A)$ for all $A \in \mathcal{A}$.

Suppose now that $E \subseteq X$ is arbitrary and $A \in \mathcal{A}$. Then, given $\varepsilon > 0$, there exist sets $I_1, I_2, \dots \in \mathcal{A}$ whose union contains A such that

$$\sum_{n=1}^{\infty} \ell(I_n) \leq m(E) + \varepsilon.$$

Noting that $A^c \in \mathcal{A}$ and so $I_n \cap A, I_n \cap A^c \in \mathcal{A}$, we thus get the inequality

$$\begin{aligned} m(E \cap A) + m(E \cap A^c) &\leq m\left(\left(\bigcup_{n=1}^{\infty} I_n\right) \cap A\right) + m\left(\left(\bigcup_{n=1}^{\infty} I_n\right) \cap A^c\right) \\ &= m\left(\bigcup_{n=1}^{\infty} (I_n \cap A)\right) + m\left(\bigcup_{n=1}^{\infty} (I_n \cap A^c)\right) \\ &\leq \sum_{n=1}^{\infty} \ell(I_n \cap A) + \ell(I_n \cap A^c) = \sum_{n=1}^{\infty} \ell(I_n) \leq m(E) + \varepsilon \end{aligned}$$

where the equality in the last line holds by additivity on the disjoint sets $I_n \cap A, I_n \cap A^c$ (and we can pad the finite union into an infinite one with infinitely many copies of $\emptyset$). The resulting inequality holds for all $\varepsilon > 0$, so $m(E \cap A) + m(E \cap A^c) \leq m(E)$. The reverse inequality is immediate by subadditivity, hence $m(E) = m(E \cap A) + m(E \cap A^c)$ for all $E \subseteq X$, i.e. $A \in \mathcal{M}$. $\square$

Problem 6. Suppose X is a set, $m: 2^X \rightarrow [0, \infty]$ is a function satisfying (2), and $\mathcal{M} \subseteq 2^X$ is as in (1). For $A \subseteq X$, let $A^c := X \setminus A$ be the complement of A in X . Show that

- (a) $m(A \cup B) \leq m(A) + m(B)$ for all $A, B \subseteq X$;
- (b) $X \in \mathcal{M}$, $A^c \in \mathcal{M}$ if $A \in \mathcal{M}$, and $A \in \mathcal{M}$ if $m(A) = 0$;
- (c) if $A, B \in \mathcal{M}$ and $E \subseteq X$, then

$$\begin{aligned} m(E) &= m(E \cap A \cap B) + m(E \cap A \cap B^c) + m(E \cap A^c \cap B) + m(E \cap A^c \cap B^c) \\ &\geq m(E \cap (A \cup B)) + m(E \cap (A \cup B)^c); \end{aligned}$$

- (d) $\mathcal{M}$ is an algebra on X (not σ -algebra yet) and

$$m\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n m(A_k)$$

for all sequences of pairwise disjoint sets $A_1, \dots, A_n \in \mathcal{M}$;

- (e) if $A_1, A_2, \dots \in \mathcal{M}$ are pairwise disjoint, then

$$m\left(E \cap \bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} m(E \cap A_n) \text{ for all } E \subseteq X, \quad m\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} m(A_n);$$

- (f) $(X, \mathcal{M}, \mu \equiv m|_{\mathcal{M}})$ is a complete measure space.

Solution. (a) Take $A_1 = A$, $A_2 = B$, and, for integers $n > 2$, $A_n = \emptyset$. Then,

$$m(A \cup B) = m\left(A \cup B \cup \bigcup_{n=3}^{\infty} \emptyset\right) \leq m(A) + m(B) + \sum_{n=3}^{\infty} m(\emptyset) = m(A) + m(B).$$

- (b) $X \in \mathcal{M}$ because for any $E \subseteq X$,

$$m(E \cap X) + m(E \cap X^c) = m(E) + m(\emptyset) = m(E).$$

If $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$ because for any $E \subseteq X$,

$$m(E \cap A^c) + m(E \cap (A^c)^c) = m(E \cap A^c) + m(E \cap A) = m(E).$$

If $A \in \mathcal{M}$ has $m(A) = 0$, then

$$m(E) \leq m(E \cap A) + m(E \cap A^c) = m(E \cap A^c) \leq m(E)$$

where the first inequality is by (a) and the equality is by $m(E \cap A) \leq m(A)$ implying $m(E \cap A) = 0$.

- (c) If $A, B \in \mathcal{M}$, then

$$\begin{aligned} m((E \cap A) \cap B) + m((E \cap A) \cap B^c) &= m(E \cap A) \\ m((E \cap A^c) \cap B) + m((E \cap A^c) \cap B^c) &= m(E \cap A^c), \end{aligned}$$

so the left-hand sides sum to $m(E \cap A) + m(E \cap A^c) = m(E)$, proving the equality. The inequality follows from $A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$, since subadditivity says that

$$m(E \cap A \cap B) + m(E \cap A \cap B^c) + m(E \cap A^c \cap B) \geq m(E \cap (A \cup B))$$

and then we can add $m(E \cap (A^c \cap B^c)) = m(E \cap (A \cup B)^c)$ to both sides.

(d) $\mathcal{M}$ contains X and is closed under complements by (b). By (c), for any $E \subseteq X$,

$$m(E) \geq m(E \cap (A \cup B)) + m(E \cap (A \cup B)^c),$$

and the reverse inequality holds by (a), so $m(E) = m(E \cap (A \cup B)) + m(E \cap (A \cup B)^c)$; This is what it means for $A \cup B$ to be in $\mathcal{M}$. Suppose now that unions of up to $n - 1$ sets in $\mathcal{M}$ are also in $\mathcal{M}$; then, if $A_1, \dots, A_n \in \mathcal{M}$, so is

$$\bigcup_{k=1}^{n-1} A_k \cup A_n,$$

so by induction $\mathcal{M}$ is closed under arbitrary finite unions.

Now, if $A, B \in \mathcal{M}$ are disjoint sets, then

$$m(A \cup B) = m((A \cup B) \cap A) + m((A \cup B) \cap A^c) = m(A) + m(B),$$

i.e. additivity holds for two disjoint sets. If we suppose that additivity holds for $n - 1$ pairwise disjoint sets in $\mathcal{M}$, then, given pairwise disjoint sets $A_1, \dots, A_n \in \mathcal{M}$,

$$m\left(\bigcup_{k=1}^n A_k\right) = m\left(\bigcup_{k=1}^{n-1} A_k\right) + m(A_n) = \sum_{k=1}^{n-1} m(A_k) + m(A_n) = \sum_{k=1}^n m(A_k).$$

By induction, the measure of the union of any finite number of pairwise disjoint sets in $\mathcal{M}$ is equal to the sum of the measures of the component sets.

(e) Suppose $A_1, A_2, \dots \in \mathcal{M}$ are pairwise disjoint. Then, for any positive integer N ,

$$m\left(E \cap \bigcup_{n=1}^{\infty} A_n\right) \geq m\left(E \cap \bigcup_{n=1}^N A_n\right).$$

I now claim that the right-hand side can be written as $\sum_{n=1}^N m(E \cap A_n)$. This follows from an induction argument: the version for two disjoint sets is shown by

$$\begin{aligned} m(E \cap (A \cup B)) &= m(E \cap (A \cup B) \cap A) + m(E \cap (A \cup B) \cap A^c) \\ &= m(E \cap A) + m(E \cap B) \end{aligned}$$

where $(A \cup B) \cap A^c = B$ because A and B are disjoint, and, assuming the result for unions of $N - 1$ pairwise disjoint sets, we get that

$$\begin{aligned} m\left(E \cap \bigcup_{n=1}^N A_n\right) &= m\left(E \cap \bigcup_{n=1}^{N-1} A_n\right) + m(E \cap A_N) \\ &= \sum_{n=1}^{N-1} m(E \cap A_n) + m(E \cap A_N) = \sum_{n=1}^N m(E \cap A_n) \end{aligned}$$

In summary, we have shown that

$$m\left(E \cap \bigcup_{n=1}^{\infty} A_n\right) \geq \sum_{n=1}^N m(E \cap A_n).$$

In the limit as N grows arbitrarily large,

$$m\left(E \cap \bigcup_{n=1}^{\infty} A_n\right) \geq \lim_{N \rightarrow \infty} \sum_{n=1}^N m(E \cap A_n) = \sum_{n=1}^{\infty} m(E \cap A_n).$$

The reverse inequality holds by countable subadditivity, so in fact the inequality is sharp. Then, we take $E = \bigcup_{n=1}^{\infty} A_n$ to get

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} m\left(\bigcup_{k=1}^{\infty} A_k \cap A_n\right) = \sum_{n=1}^{\infty} m(A_n).$$

(f) Suppose $A_1, A_2, \dots \in \mathcal{M}$. If $m(E \cap \bigcap_{n=1}^{\infty} A_n) = \infty$, then by the assumption (2) we have $m(E) = \infty$ as well, in which case $m(E \cap \bigcup_{n=1}^{\infty} A_n) + m(E \cap \bigcap_{n=1}^{\infty} A_n^c) = m(E)$. Assume now that $m(E \cap \bigcap_{n=1}^{\infty} A_n) < \infty$. Define $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n > 1$, and note that since $\mathcal{M}$ is an algebra, $\bigcup_{k=1}^n A_n = \bigcup_{k=1}^n B_n \in \mathcal{M}$ for all $n \in \mathbb{N}$. By (e), since the B_n are pairwise disjoint,

$$\begin{aligned} m\left(E \cap \bigcup_{n=1}^{\infty} A_n\right) + m\left(E \cap \bigcap_{n=1}^{\infty} A_n^c\right) &= m\left(E \cap \bigcup_{n=1}^{\infty} B_n\right) + m\left(E \cap \bigcap_{n=1}^{\infty} B_n^c\right) \\ &= \sum_{n=1}^{\infty} (m(E \cap B_n)) + m\left(E \cap \bigcap_{n=1}^{\infty} B_n^c\right) \end{aligned}$$

The sum is finite, so given $\varepsilon > 0$ there exists N such that $\sum_{n=1}^{\infty} m(E \cap B_n) < \varepsilon + \sum_{n=1}^N m(E \cap B_n)$. Thus

$$\begin{aligned} \sum_{n=1}^{\infty} (m(E \cap B_n)) + m\left(E \cap \bigcap_{n=1}^{\infty} B_n^c\right) &\leq \sum_{n=1}^N (m(E \cap B_n)) + m\left(E \cap \bigcap_{n=1}^N B_n^c\right) + \varepsilon \\ &= m\left(E \cap \bigcup_{n=1}^N B_n\right) + m\left(E \cap \bigcap_{n=1}^N B_n^c\right) + \varepsilon = m(E) + \varepsilon \end{aligned}$$

where the first equality follows from part (e), since we can pad the union of the (pairwise disjoint) sets $B_1, \dots, B_N$ into a countable union with copies of $\emptyset$; and the second equality follows from $\bigcup_{k=1}^n B_n$ being in $\mathcal{M}$ as stated earlier. In summary,

$$m\left(E \cap \bigcup_{n=1}^{\infty} A_n\right) + m\left(E \cap \bigcap_{n=1}^{\infty} A_n^c\right) \leq m(E) + \varepsilon$$

for every $\varepsilon > 0$, so the left-hand side is bounded above by $m(E)$. Yet again the reverse inequality is true by subadditivity, so $m(E \cap \bigcup_{n=1}^{\infty} A_n) + m(E \cap \bigcap_{n=1}^{\infty} A_n^c) = m(E)$ for all E , which is what it means for the union $\bigcup_{n=1}^{\infty} A_n$ to be in $\mathcal{M}$; this proves that $\mathcal{M}$ is a σ -algebra.

To show that $\mu \equiv m|_{\mathcal{M}}$ is a measure, we have $\mu(\emptyset) = 0$ by assumption and if $A_1, A_2, \dots \in \mathcal{M}$ are pairwise disjoint, then by (e)

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

This completes the proof that $(X, \mathcal{M}, \mu)$ is a measure space.

Finally, this measure space is complete: given $A \in \mathcal{M}$ with $\mu(A) = 0$, any subset $B \subseteq A$ has $m(B) \leq m(A) = \mu(A)$, which implies $m(B) = 0$, so by part (b), $B \in \mathcal{M}$. $\square$